ProprioVLA: Self-Supervised Proprioceptive Grounding for Vision-Language-Action Policies

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Abstract

Vision-language-action (VLA) policies have shown strong promise as generalist controllers for robot manipulation, yet their action predictions can remain fragile under changes in visual perspective. We identify a systematic grounding failure: lateral camera shifts induce co-directional shifts in predicted robot trajectories, suggesting that current VLAs may exploit image-space shortcuts, such as pixel-coordinate correlations, rather than grounding actions in proprioceptive state and robot-object geometry. To bridge this gap, we introduce PROPRIOVLA, a self-supervised learning framework that incentivizes proprioceptive grounding in VLA policies. PROPRIOVLA guides the policy to visually attend to the robot itself by conditioning visual queries on proprioceptive state, based on the key insight that state-aligned residual flow provides implicit supervision for self-grounding. Specifically, PROPRIOVLA first mines confidence-weighted self-grounding heatmaps from robot interaction by selecting residual motion that is salient in the image and aligned with proprioceptive change. It then optimizes two complementary objectives: a visual self-grounding objective that aligns state-conditioned attention with these heatmaps, and a proprioceptive reconstruction objective that ties the attended representation back to the robot's own state. This yields self-supervised proprioceptive grounding without manual annotations, external grounding datasets, or stronger foundation-model supervision. Experiments on both real-robot and simulation environments show that PROPRIOVLA improves task success and robustness over action-supervised VLA adaptation, suggesting that proprioceptive grounding is a practical route toward spatially reliable VLA control.

1 Introduction

Vision-language-action (VLA) policies aim to map language instructions and visual observations to robot actions, making them a promising interface for generalist manipulation. Recent robot foundation models show that broad pretraining and large robot datasets can yield policies that transfer across tasks and embodiments [2, 3, 7, 5]. For manipulation, however, success depends on more than recognizing objects in the image: the policy must relate visual evidence to the robot's own proprioceptive state and to the geometry of the surrounding scene. Spatial robustness under changes in visual perspective is therefore a core requirement for reliable VLA control.

Despite this progress, we find that fine-tuned VLA policies can remain weakly grounded in their own embodiment. In controlled camera-perturbation evaluations where the underlying robot-scene state is unchanged, lateral camera shifts induce co-directional shifts in predicted robot trajectories. This systematic failure suggests that current VLAs may exploit image-space shortcuts, such as pixel-coordinate correlations, rather than grounding actions in proprioceptive state and robot-object geometry. Such shortcuts are especially problematic during fine-tuning, where a policy may match

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demonstrations under a fixed camera distribution while remaining brittle when the same physical state is viewed from a different perspective.

To address this issue, we introduce PROPRIOVLA, a self-supervised learning framework for proprioceptive grounding in VLA policies. The central idea is to guide the policy to visually attend to the robot itself by conditioning visual queries on proprioceptive state. Rather than treating the state as the supervision signal, PROPRIOVLA uses state as a guiding input and obtains supervision from state-aligned residual flow. This signal captures residual visual motion that is both locally salient in the image and synchronized with proprioceptive change, revealing where the robot's own visual evidence appears during interaction.

PROPRIOVLA turns this signal into confidence-weighted self-grounding heatmaps and uses them to train two complementary objectives. The heatmaps are mined from robot interaction by selecting residual motion that is salient in the image and aligned with proprioceptive change, abstaining from supervision when the signal is uncertain. They supervise a visual self-grounding objective that aligns state-conditioned attention with robot-related visual evidence. A complementary proprioceptive reconstruction objective requires the attended representation to recover the robot's own state, encouraging the grounded visual representation to remain tied to proprioception. Together, these objectives incentivize proprioceptive grounding without manual annotations, external grounding datasets, or supervision from stronger foundation models.

We evaluate PROPRIOVLA in both real-robot and simulation environments, including controlled camera-perturbation settings that directly test spatial robustness. Across these evaluations, PROPRIOVLA improves task success and robustness over action-supervised VLA adaptation. Ablations further support the role of both components: self-grounding heatmaps provide spatial supervision for state-conditioned attention, while proprioceptive reconstruction anchors the attended representation to the robot state. These results suggest that self-supervised proprioceptive grounding is a practical route toward spatially reliable VLA control.

Our contributions are threefold:

- We identify a systematic grounding failure in VLA policies: camera shifts can induce co-directional shifts in predicted robot trajectories, revealing reliance on image-space shortcuts rather than proprioceptive grounding.
- We introduce state-aligned residual flow as an implicit supervision signal for self-grounding, enabling confidence-weighted heatmaps to be mined from robot interaction without manual annotations, external grounding datasets, or stronger foundation-model supervision.
- We propose PROPRIOVLA, a self-supervised training framework that couples state-conditioned visual attention with proprioceptive reconstruction to improve task success and robustness in real-robot and simulation environments.

2 Related Work

Vision-language-action models. Vision-language-action (VLA) policies scale language-conditioned robot control by coupling vision-language backbones with robot action prediction. RT-1 and RT-2 established the value of large robot datasets and web-scale semantic priors [2, 3]; recent open and generalist models further improve transfer across tasks, embodiments, and action parameterizations [7, 5, 1]. However, this line of work mainly optimizes task success under the training observation distribution. It does not directly test whether the action-relevant visual evidence is anchored to camera-invariant robot-object geometry. PROPRIOVLA studies this missing failure mode: under camera perturbations, a fine-tuned VLA can follow image-coordinate shortcuts instead of grounding actions in the robot's own state.

Spatial and proprioceptive grounding in VLA policies. Recent work improves VLA perception by making space, viewpoint, or proprioception more explicit. TraceVLA adds visual trajectory prompts to increase spatial-temporal awareness [11]; SpatialVLA injects 3D spatial representations [8]; OC-VLA predicts actions in camera space to reduce observation-action mismatch [9]; and VistaBot uses geometry-aware view synthesis for cross-view robustness [4]. These methods show that perception is a central bottleneck, but they depend on explicit prompts, 3D encodings, camera coordinate transforms, or generative view modules. A complementary line questions proprioception itself: State-

free Policy removes proprioceptive inputs to avoid trajectory overfitting [10], while GAP shows that proprioceptive gradients can suppress visual learning during motion transitions [6]. PROPRIOVLA takes a different stance: it does not discard proprioception or treat it as a passive token. Instead, it uses proprioceptive change to mine state-aligned residual-flow heatmaps, turning the robot's own motion into self-supervision for visual attention and proprioceptive reconstruction.

3 Does a VLA Know Where It Is?

A manipulation policy must know where it is, not only what object the language instruction names. We probe this self-location requirement at two levels. The action-level probe asks whether a predicted trajectory remains tied to the fixed robot-scene state when only the camera view is shifted. The representation-level probe asks whether robot-relevant attention localizes the arm, gripper, or end-effector rather than only task objects or background context. Together, these probes expose a *missing-self* failure mode: a policy can appear task-grounded while lacking a stable visual grounding of the body that acts.

3.1 Finding 1: The Action Frame Is Entangled with the Image Frame

We diagnose action-frame grounding by perturbing the view while holding the physical manipulation state fixed. Given observation o_t, instruction x, proprioceptive state s_t, and policy π_θ, let $\tau_\theta = \pi_\theta(o_t, x, s_t)$ be the predicted trajectory from the original view. We then apply a lateral camera or image shift $T_{\Delta v_{\mathrm{cam}}}$ and compute $\tau'_\theta = \pi_\theta(T_{\Delta v_{\mathrm{cam}}}(o_t), x, s_t)$. The instruction, robot state, object configuration, and physical scene are unchanged; only the image frame moves.

If the action frame were grounded in robot-scene geometry, the prediction would remain stable under this camera-only perturbation, up to the known transform used for visualization. A missing-self failure appears when the predicted end-effector displacement follows the image displacement. We quantify this effect after projecting the trajectory change onto a common lateral coordinate system:

$$\mathrm{TSR} = \frac{\|\Delta\tau_{\mathrm{pred}}\|_2}{\|\Delta v_{\mathrm{cam}}\|_2 + \epsilon},$$
$$\mathrm{SDA} = \frac{\langle \Delta\tau_{\mathrm{pred}}, \Delta v_{\mathrm{cam}} \rangle}{\|\Delta\tau_{\mathrm{pred}}\|_2 \|\Delta v_{\mathrm{cam}}\|_2 + \epsilon}.$$

High TSR means the prediction changes strongly under a view-only shift; high SDA means the change is co-directional with the image shift. The combination indicates that the learned action frame is partially entangled with the image frame rather than fully anchored to proprioceptive state and robot-object geometry.

3.2 Finding 2: Task Grounding Does Not Imply Self-Grounding

We next test whether action-relevant representations localize the acting body. For a VLA or its vision-language backbone, we use robot-centric prompts such as "Where is the robot arm?", "Please identify the robot itself in the scene," and "Where is the manipulator?" We then visualize attention heatmaps, attention rollout, Grad-CAM maps, cross-attention maps, or token relevance maps. The same analysis can be applied to policy tokens used for action prediction, comparing task relevance with self relevance.

This probe separates task grounding from self-grounding. A model may attend to task objects, the tabletop, or salient background regions while assigning little relevance to the robot arm, gripper, or end-effector. Such attention can support object recognition and instruction following, but it does not bind the action to the body that will execute it. Robust VLA control therefore requires binding proprioceptive state to visual evidence of the acting body, rather than treating proprioception as a passive input that the visual pathway can ignore.

4 PROPRIOVLA

PROPRIOVLA adds a proprioceptive grounding path to a VLA policy while leaving the standard action-supervised objective intact. During training, the method mines self-grounding targets from

state-aligned residual motion, uses proprioceptive state as a query over visual features, and anchors the queried visual representation by reconstructing proprioceptive information. At test time, only the learned state-conditioned attention path is used; the mined heatmaps are not required.

4.1 Self-Grounding from State-Aligned Motion

The robot reveals itself through motion that is synchronized with its state change. Given consecutive observations (o_t, o_{t+1}) and proprioceptive states (s_t, s_{t+1}), we define $\Delta s_t = s_{t+1} - s_t$. We estimate dense visual motion $F_t(p)$ between frames and subtract global or background motion to obtain residual motion $R_t(p)$ at image location p. This residual suppresses camera drift and static-scene variation, emphasizing motion caused by the robot or by objects directly coupled to the robot.

We convert residual motion into a confidence-weighted self-grounding target. Let $\mathrm{sal}(R_t(p))$ denote local residual-motion saliency and let

$$\alpha_t(p) = \big[\cos\big(\psi_R(R_t(p)), \psi_s(\Delta s_t)\big)\big]_+$$

measure state-motion alignment after lightweight projections ψ_R and ψ_s. The unnormalized heatmap score is

$$e_t(p) = \mathrm{sal}(R_t(p))\, \alpha_t(p).$$

We normalize e_t over the image or patch grid to obtain H_t, and set a confidence $c_t \in [0, 1]$ from the total aligned-motion mass. Low confidence downweights frames where the self-motion cue is weak or ambiguous. Thus the training target is mined from interaction alone, without manual masks, external grounding datasets, or stronger foundation-model supervision.

4.2 Proprioception as a Visual Query

In PROPRIOVLA, proprioception determines where the policy looks rather than merely adding another token to the transformer. We encode the proprioceptive state as

$$q_t^s = f_s(s_t),$$

and obtain visual keys and values $K_t, V_t = f_v(o_t)$ from the VLA vision encoder. The state query attends over visual patches according to

$$A_t = \mathrm{softmax}\left(\frac{q_t^s K_t^\top}{\sqrt{d}}\right), \qquad z_t^{\mathrm{self}} = A_t V_t.$$

Here A_t is a distribution over image patches, and z_t^{self} is the self-grounded visual feature passed to the action head together with the policy's standard visual-language representation.

The mined heatmap supervises this state-conditioned attention through

$$\mathcal{L}_{\mathrm{ground}} = c_t\, D(A_t, H_t),$$

where D is cross-entropy or KL divergence after resizing H_t to the attention grid. This loss turns proprioception into an active grounding mechanism: the state embedding selects image evidence corresponding to the current state. At inference time, the learned query directly produces A_t and z_t^{self} from the current observation and state.

4.3 Proprioceptive Anchoring by Reconstruction

The self-grounding loss specifies where to attend, but it does not guarantee what the attended feature encodes. We therefore add a reconstructive proprioceptive anchoring objective. The reconstruction path is

$$s_t \to q_t^s \to A_t \to z_t^{\mathrm{self}} \to \hat{s}_t,$$

where the attended representation is explicitly

$$z_t^{\mathrm{self}} = \sum_p A_t(p) V_t(p), \qquad \hat{s}_t = g_s(z_t^{\mathrm{self}}).$$

The state reconstruction loss is

$$\mathcal{L}_{\mathrm{state}} = \|\hat{s}_t - s_t\|_2^2.$$

For paired observations, we also reconstruct state change:

$$\widehat{\Delta s_t} = g_\Delta(z_t^{\text{self}}, z_{t+1}^{\text{self}}), \qquad \mathcal{L}_{\text{motion}} = \|\widehat{\Delta s_t} - \Delta s_t\|_2^2.$$

The full reconstruction objective is

$$\mathcal{L}_{\text{rec}} = \mathcal{L}_{\text{state}} + \beta \mathcal{L}_{\text{motion}},$$

and the complete training loss is

$$\mathcal{L} = \mathcal{L}_{\text{act}} + \lambda_g \mathcal{L}_{\text{ground}} + \lambda_r \mathcal{L}_{\text{rec}}.$$

Here $\mathcal{L}_{\text{act}}$ is the standard imitation loss used to fine-tune the VLA. The reconstruction heads receive only z_t^{self}, never the raw state s_t or state embedding q_t^s. This implementation detail prevents trivial copying and makes reconstruction a visual anchoring regularizer. The policy is therefore trained not only to look at the robot, but also to encode the robot's state in the visual feature it attends to.

5 Experiments

The experiments test whether proprioceptive self-grounding improves spatial reliability beyond ordinary action supervision and state concatenation. The evaluation links three evidence types: task performance, action stability under camera perturbations, and visual grounding of the acting robot.

5.1 Experimental Setup

We evaluate a VLA backbone fine-tuned on language-conditioned manipulation demonstrations in simulation and on a real robot. Each episode contains visual observations o_t, proprioceptive state s_t, language instruction x, and action supervision a_t. For robustness evaluation, we keep the physical robot-scene state, instruction, and proprioceptive state fixed, then apply lateral view shifts of controlled magnitude. This isolates camera-induced changes from changes in the underlying manipulation state.

We compare four policies under the same demonstrations, action labels, and view-shift protocol: Base VLA, action-supervised fine-tuning, VLA with proprioception concatenation, and PROPRIOVLA. We report task success rate, action prediction error, Trajectory Shift Ratio, Shift Direction Agreement, and a grounding score. The grounding score is computed as either attention-heatmap alignment with H_t or attention mass on robot regions. Lower trajectory shift and direction agreement indicate a more stable action frame; higher grounding score indicates stronger localization of the acting body. Each setup reports the task suite, demonstration count, train/test split, camera shifts, action parameterization, proprioceptive normalization, optimizer, and loss weights λ_g, λ_r, and β.

5.2 Simulation Experiments

Simulation provides controlled access to camera shifts, object layouts, and ground-truth robot geometry. We evaluate normal, mild-shift, medium-shift, and large-shift settings. For each setting, we report task success, action prediction error, Trajectory Shift Ratio, Shift Direction Agreement, and grounding score. Robustness curves plot these metrics against shift magnitude, making the co-movement between predicted actions and image shifts directly visible.

We also evaluate held-out object layouts and camera positions. These splits separate memorization of a fixed training camera from grounding that transfers across views. Qualitative trajectory overlays compare the original prediction with shifted-view predictions after mapping them to the same robot-centric coordinate frame.

5.3 Real-World Experiments

The real-robot evaluation tests whether the same failure mode appears under sensor noise, calibration error, and real scene variation. We evaluate several manipulation tasks under the original view and shifted views, reporting task success, success drop under perturbation, action prediction error on logged demonstrations, and grounding score.

For qualitative analysis, we visualize baseline attention, the mined state-aligned heatmap, PROPRIOVLA attention, and trajectory overlays for the same scene. The comparison checks whether attention remains dominated by task objects and background context or shifts toward the arm, gripper, and end-effector while the trajectory remains stable.

5.4 Ablations

The ablation study isolates the mechanisms that distinguish PROPRIOVLA from simple state conditioning. We evaluate four variants: (1) removing the visual self-grounding loss $\mathcal{L}_{\text{ground}}$; (2) removing the reconstruction loss $\mathcal{L}_{\text{rec}}$; (3) using raw optical flow instead of state-aligned residual motion; and (4) replacing the proprioceptive visual query with state concatenation. Each ablation is evaluated with the same success, action, trajectory-shift, and grounding metrics.

These ablations test the central mechanism: the policy must use proprioception to ground visual attention and anchor the attended feature, rather than merely receive proprioceptive values as extra inputs. The comparison separates three alternatives: gains from additional state, gains from motion saliency alone, and gains from explicit proprioceptive self-grounding.

6 Limitations

PROPRIOVLA assumes that the robot is at least partially visible during training. If the embodiment is consistently outside the camera view, the state-aligned motion signal cannot provide useful self-grounding supervision, and the visual query has no direct evidence from which to recover the robot's own state.

State-aligned residual motion can be ambiguous when non-robot motion is coupled to the robot. Grasped objects, deformable materials, moving backgrounds, or other agents may move synchronously with proprioceptive change and receive high alignment scores. Confidence weighting reduces weak supervision, but it cannot fully distinguish the robot body from every object moved by that body.

Finally, the evaluation covers a finite set of embodiments, camera perturbations, and manipulation tasks. Broader multi-camera setups, mobile manipulation, multi-robot scenes, long-horizon tasks, and highly dynamic environments remain important tests for whether proprioceptive self-grounding scales beyond controlled perturbation regimes.

7 Conclusion

This paper studies a missing-self failure mode in VLA policies: under camera-only viewpoint shifts, predicted robot trajectories can move with the image frame even when the physical robot-scene state is unchanged. The diagnostic shows that task grounding and self-grounding are distinct requirements for spatially reliable manipulation.

PROPRIOVLA addresses this issue through three connected mechanisms: state-aligned motion supervision, proprioception as a visual query, and proprioceptive anchoring by reconstruction. The framework mines self-grounding heatmaps from robot interaction without manual masks or external grounding datasets, uses proprioceptive state to select robot-related visual evidence, and requires the attended visual feature to recover the robot's own state. Learning to see the self is a practical step toward VLA policies whose actions are grounded in embodiment rather than in the image frame alone.

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